

Reviewer Pack — Minimal Order of Quadratic Forms Bounds Communication Comple...

Entry f20f24a2bf8b · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-29 19:28:34 UTC

Minimal Order of Quadratic Forms Bounds Communication Complexity for XOR

Entry ID: f20f24a2bf8b **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 19:28:34 UTC

1. Conjecture

Field A (mathematical branch): Algebra (Quadratic Forms) **Field B** (complexity object): Boolean Function Complexity: Circuit Complexity of XOR

Statement:

For any given n -variable XOR Boolean function f , the minimal order $q(f)$ of a quadratic form that correctly classifies f is related to the circuit complexity $C(f)$ as follows: $O(q(f)) \leq C(f) \leq \Omega(q(f)^2)$. For all instances with property P , the following holds: if there exists a polynomial-time computable quadratic form g such that $g[f] = 1$ for all x in $\{0, \dots, n-1\}$, then $q(f) \leq |\text{support}(g)|$.

Rationale (proposer's reasoning):

Quadratic forms have been used in real algebraic geometry and may provide a computational link to the circuit complexity of XOR Boolean functions. The minimal order of a quadratic form could reveal hidden structure within the function that is not apparent through traditional circuit analysis.

Taxonomy category: cg_kw_andreev (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ee54ba7a51b200da

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given XOR Boolean function f with n variables, if there exists a polynomial-time computable quadratic form g such that $|\text{support}(g)| \geq n/2$ and no seed produces an aggregate metric value > 10 across all seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.90	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	SAFE
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=1.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_xor_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def support_size(g, f):
        return sum(1 for x in range(len(f)) if g(x) == 1 and f[x] == 1)

    def quadratic_form_complexity(g):
        # Placeholder for actual computation
        return len(g)

    def circuit_complexity(f):
        # Placeholder for actual computation
        return len(f)

    n = random.choice([5, 10, 15, 20, 30, 40])
    f = generate_xor_function(n)
    g = lambda x: sum(x[i] * i for i in range(n)) % 2

    q_f = quadratic_form_complexity(g)
    C_f = circuit_complexity(f)

    if q_f == 0:
        return {
            "metric_name": "q(f)",
            "metric_value": q_f,
            "instances_tested": 1,
            "n_max": n,
            "conjecture_holds": False,
            "counterexample": "mapping_undefined"
        }
```

```

0_qf = math.log2(q_f)
0omega_qf_squared = q_f ** 2

if 0_qf > C_f or C_f < 0omega_qf_squared:
    return {
        "metric_name": "q(f)",
        "metric_value": q_f,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": f"0(q(f)) > C(f) or C(f) < Ω(q(f)^2)"
    }

if support_size(g, f) < n / 2:
    return {
        "metric_name": "q(f)",
        "metric_value": q_f,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": f"support_size(g) < n/2"
    }

return {
    "metric_name": "q(f)",
    "metric_value": q_f,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": True,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=insufficient_support_fraction")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5bb380ed.py", line 89, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5bb380ed.py", line 39, in run_trial
    q_f = quadratic_form_complexity(g)
          ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5bb380ed.py", line 29, in quadratic_for
    return len(g)
           ~~~~~^
TypeError: object of type 'function' has no len()
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means the pre-registered support condition could not be unambiguously met. | next: Investigate and fix the crash in the test code to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15121
2	preregistration	ollama_remote	glm4:latest	0	0	12413
3	novelty	ollama_remote	glm4:latest	0	0	11583
4	novelty	ollama_remote	glm4:latest	0	0	8680
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18954
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11043
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11272
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	36187
9	judge	ollama_remote	glm4:latest	0	0	23795

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 149048 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/f20f24a2bf8b.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/f20f24a2bf8b.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/f20f24a2bf8b.tar.gz` (if generated)